Skills Summary

Diagnosing Separation and Initial-Condition Errors

Use this reference while completing your worksheet or when studying.

Skill 1 — Integrating both sides

When the right side contains no y, the equation is already separated: integrate straight across.

$$\frac{dy}{dx} = f(x) \implies y = \int f(x)\,dx + C$$

One constant is enough — a C on each side collapses into their difference. The $+C$ is not decoration: it is the entire family of solution curves, and the initial condition's only job is to choose one of them.

Example: Solve $\dfrac{dy}{dx} = 8x^3 - 3$.

Step 1. The right side has no y in it, so no separation step is needed — this is a plain antiderivative question.

Step 2. Integrate term by term: $\displaystyle\int (8x^3 - 3)\,dx$, giving $\boldsymbol{y = 2x^4 - 3x + C}$. Check by differentiating back.

Try it: Solve $\dfrac{dy}{dx} = 10x - 6x^2$.

check: $\boldsymbol{y = 5x^2 - 2x^3 + C}$

Skill 2 — Separating the variables

Rule: you may integrate only when *every* y sits with dy and *every* x sits with dx:

$$\frac{dy}{dx} = f(x)\,g(y) \implies \frac{1}{g(y)}\,dy = f(x)\,dx \implies \int \frac{dy}{g(y)} = \int f(x)\,dx.$$

y is a *function* of x, so a stray y on the dx side can never be treated as a constant. Two integrals worth memorising: $\displaystyle\int \frac{dy}{y} = \ln|y|$ and $\displaystyle\int y^{-2}\,dy = -\frac{1}{y}$ (the minus sign is lost constantly).

Example: $\dfrac{dy}{dx} = \dfrac{x}{y}$, $y(0) = 4$. Find $y(3)$.

Step 1. There is a y on the right, so nothing may be integrated until it is moved across — multiply both sides by y first.

Step 2. $y\,dy = x\,dx$ gives $\frac{y^2}{2} = \frac{x^2}{2} + C$. Apply $y(0) = 4$ here, in the form you derived: $C = 8$, so $y^2 = x^2 + 16$ and $y = \sqrt{x^2 + 16}$; then $y(3) = \sqrt{9 + 16} = \mathbf{5}$.

Try it: $\dfrac{dy}{dx} = \dfrac{x}{y}$, $y(0) = 12$. Find $y(5)$.

check: **13**

(!) Watch out: never square-root term by term. $\sqrt{x^2+16} \neq x+4$. Apply the initial condition to the equation you actually derived, before any rearranging that changes what C means.

Skill 3 — Where the constant of integration goes

The exponential trap. Solving $\dfrac{dy}{dx} = ky$ gives $\ln|y| = kx + C$. Exponentiating turns the sum into a **product**:
$|y| = e^{kx+C} = e^C e^{kx} \implies y = Ae^{kx}$, where $A = e^C$.
So the constant is a **multiplier**, never a term: $y = e^{kx} + C$ is not a solution for any $C \neq 0$. With $y(0) = y_0$ the answer is always $y = y_0 e^{kx}$.
One-line safety check: differentiate your answer and compare it with the original right-hand side.

Example: $\dfrac{dy}{dx} = 5y$, $y(0) = 7$. Find $y(0.2)$.
Step 1. The rate is proportional to the amount, so the solution is exponential and the constant must be carried through as a factor, not added afterwards.
Step 2. $y = Ae^{5x}$ with $y(0) = A = 7$, so $y = 7e^{5x}$ and $y(0.2) = 7e^1 =$ **19.03**. Check: $y' = 35e^{5x} = 5y$.

Try it: $\dfrac{dy}{dx} = 2y$, $y(0) = 6$. Find $y(0.5)$.
check: **16.31**